

QM640 - Data Analytics Capstone

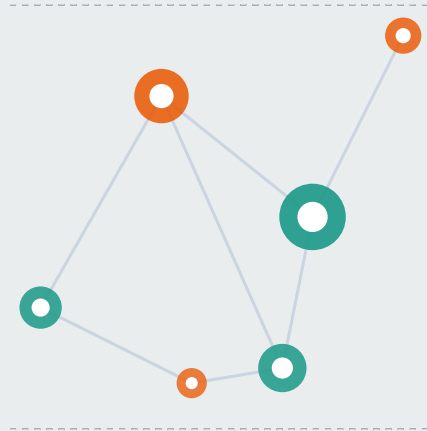


Municipal Correlates of Licensed 5G NR Infrastructure in Brazil:

Presence, Intensity, and Digital-Infrastructure Profiles

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Executive Summary



Objective

This study classifies Brazilian municipalities by 5G NR (New Radio) deployment, identifies predictors of NR and SLP - Limited Private Service station density and examines structural correlates of station intensity.

Importance

Creates a national screening framework distinguishing where 5G appears versus its deployment intensity.

Data

Merged Anatel - Brazilian Telecommunications Agency and IBGE - Brazilian Institute of Geography and Statistics datasets with demographic and infrastructure metrics for all 5,571 Brazilian municipalities.

Results

Population scale strongly predicts initial deployment presence, but structural factors only weakly explain per-capita intensity, and SLP activity is primarily linked to municipal wealth and legacy telecom infrastructure.

Usage

Serves as a reproducible prioritization and benchmarking tool for Telecom regulators. Helps MNOs - Mobile Networks operators prioritize rollouts and reduce broad survey efforts and provides enterprise a preliminary private network feasibility.

Gap Analysis



Gist of Literature Review

Deployment Economics

Infrastructure rollout heavily favors areas with high population scale and density to absorb steep fixed costs, while coverage obligations are deployed to prevent lower-density markets from being left behind.

Inherited Advantage

Brazilian connectivity literature demonstrates that access persistently concentrates in wealthier localities with established legacy foundations (fiber backhaul and LTE), reinforcing macro-regional divides.

Presence vs. Intensity

Deciding to enter a municipality (extensive margin) is economically distinct from scaling station capacity post-entry (intensive margin), necessitating two-part hurdle and count modeling rather than a single uniform regression.

Gap Analysis



Justification of the Problem

Empirical Gap in 5G Research

Prior Brazilian municipal research focused primarily on fixed broadband, individual survey access, or selected metropolises prior to 5G; no comprehensive study covered the entire 5G-era frame across all 5,571 municipalities.

Conflation of Scale & Intensity

Existing literature does not resolve whether socioeconomic and inherited infrastructure drivers merely predict market entry (scale effect) or genuinely explain continuous per-capita deployment density once present.

Nationwide Benchmarking

A nationwide, methodologically rigorous baseline is required to provide actionable prioritization data for regulators, mobile carriers, and enterprises in Brazil.

Research Questions



RQ1: Rollout Classification

Can structural municipal characteristics predict whether a Brazilian municipality is a high-count licensed 5G NR rollout site, and does this hold on a per-capita basis?

Hypothesis

Null (H₀): Random forest discriminates high-count municipalities no better than chance (ROC-AUC = .50).

Alternative (H_a): Held-out ROC-AUC exceeds .50.

Statistical Method

Random forest classifier with 5-fold cross-validated grid tuning, state-grouped cross-validation, and permutation feature importance.

Test Results

Ho was rejected. The model effectively discriminated high raw counts, but performance dropped to on a per-capita basis, indicating population scale drove much of the original signal.

Research Questions



RQ2: Structural Correlates

To what extent are structural characteristics associated with population-normalized NR station deployment intensity, and which carry statistical significance?

Hypotheses

Ho-P (Predictive Null): No held-out predictive value (bootstrap 95% CI for R^2 includes zero).

Ha-P (Predictive Alt): Has held-out predictive value (bootstrap 95% CI excludes zero).

Ho-I (Inferential Null): No coefficient is different from zero in the zero-truncated model.

Ha-I (Inferential Alt): At least one characteristic has a non-zero coefficient.

Statistical Method

This hurdle framework handles zero-inflated data by modeling initial 5G market entry via logistic regression, predicting per-capita intensity in deployed locations with a zero-truncated negative binomial model, and auditing predictive power using cross-validated ridge regression.

Test Results

Ho-I was rejected; Ho-P was not. Presence was moderately predictable, but intensity was weakly explained. Population density, fiber, and GDP showed modest positive effects, while station counts grew sub-proportionally with population.

Research Questions



RQ3: Digital-Infrastructure Profile Segmentation

How do municipalities cluster on digital-infrastructure profile indicators, and how strongly is cluster membership associated with geographic region?

Hypotheses

Null (H₀): Cluster membership is independent of macro-region.

Alternative (H_a): Cluster membership and region are associated.

Statistical Method

K-means clustering (evaluated via silhouette coefficient and bootstrap adjusted Rand index) combined with a Chi-square test of independence and Cramér's V to evaluate regional association.

Test Results

Ho was rejected. The k-means algorithm identified two stable but softly separated infrastructure profiles. The Chi-square test confirmed an association between cluster membership and region, though Cramér's V demonstrated that this relationship was modest.

Research Questions



RQ4: SLP Station Intensity (Exploratory)

Which municipal characteristics are associated with high private-network-related SLP station intensity, and does the association generalize across states?

Hypotheses

Null (H₀): Structural characteristics are jointly unrelated to the likelihood of high SLP intensity.

Alternative (H_a): At least one characteristic is significantly associated with high SLP intensity.

Statistical Method

Balanced logistic regression with state-clustered standard errors (including a $t(26)$ small-cluster correction), likelihood-ratio tests, and multiple target aggregation sensitivity checks.

Test Results

H₀ was rejected. The model attained a held-out ROC-AUC of .810, showing high SLP intensity was significantly associated with higher wealth, population size, LTE access, and fiber intensity. However, performance dropped under state-grouped validation, indicating geographic heterogeneity.

Data Description and EDA



Dataset Scope

Sources & Coverage

Official public records sourced from IBGE and Anatel, covering exactly 5,571 unique rows to represent all Brazilian municipalities.

Data Quality

Data missingness is analytically immaterial, reaching a maximum rate of only 0.11% across the entire frame.

Columns & Features

42 Engineered Features

Broad spectrum of variables supporting robust profile analysis:

- **Demographics:** Population and density.
- **Economy:** GDP per capita.
- **Network:** Fiber intensity and legacy LTE.
- **Speeds:** Exploratory download metrics.

Target Footprint

Infrastructure Tracking

Analyzed and monitored through precise operational metrics:

5G New Radio (NR)

Tracked via active licensed station counts across regions.

Private Networks

Regulated through official Serviço Limitado Privado (SLP) authorizations.



Data Dictionary (sample)

Variable	Definition	Type	Role
MUNICIP_ID	7-digit IBGE municipal code (string)	Key	Join key
POP_2024	Estimated Resident population, 2024	Continuous	Predictor (RQ1-RQ4)
GDP_PER_CAP	Municipal GDP per capita (BRL)	Continuous	Predictor (RQ1-RQ4)
FIBER_PER_100	Fiber broadband accesses per 100 inhabitants	Continuous	Predictor (RQ1-RQ4)
NR_STATION_CNT	Licensed 5G NR radio stations (advanced footprint)	Integer	RQ1 Target / RQ2 Count
HIGH_NR_P75	1 if NR_STATION_CNT > 9 (75th percentile)	Binary	RQ1 Primary Target
SLP_PER_100K_P	SLP stations per 100,000 population	Continuous	RQ4 Target Construction
HIGH_SLP	1 if SLP_PER_100K_P > 1,618 (75th percentile)	Binary	RQ4 Target

Data Description and EDA

Skewness & Zero-Inflation

Municipal data exhibits severe right-skewness across density, GDP, and NR counts. 60.7% of all municipalities report exactly zero licensed NR stations, justifying a two-part hurdle model.

Correlation Structure

Raw station counts are heavily dominated by municipal scale effects (correlation with population), validating the decision to engineer per-capita target intensities.

Speed by Rollout Stage

ANOVA test confirms variations: high-count NR municipalities (>9 stations) achieve 55.63 Mbps median speed vs 36.60 Mbps in zero-footprint areas.

Regional Disparities

The South region leads Brazil with 46.8% of municipalities having at least one licensed NR station, while the Center-West lags behind with the lowest rate at 33.3%.

Fiber vs. NR Density

Infrastructure Constraint

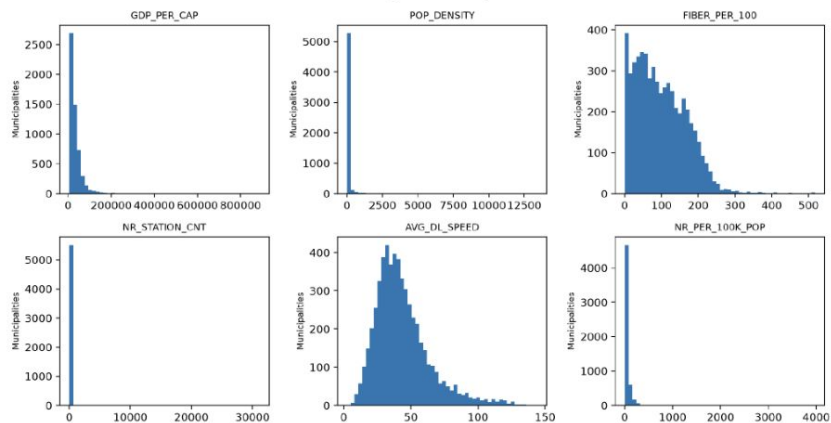
Analyzing only NR-present municipalities reveals that while fiber intensity is a necessary contextual factor, it is entirely insufficient on its own to explain the density of 5G deployment.

This underscores the complexity of regional rollouts beyond basic backhaul availability.

Data Description and EDA

Distribution Diagnostics for Key Variables

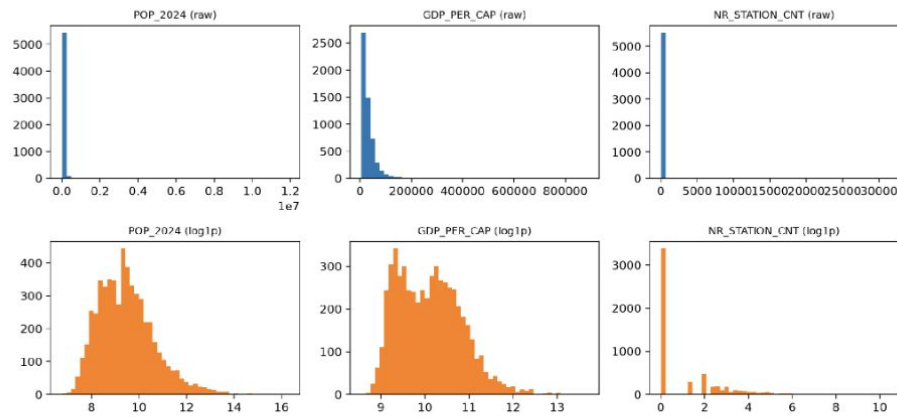
Distribution diagnostics for key variables



Note. The distributions reveal strong right skew in population density, GDP per capita, fiber intensity, NR station counts, and population-normalized station density.

Skewed Variables on Raw Versus log1p Scales

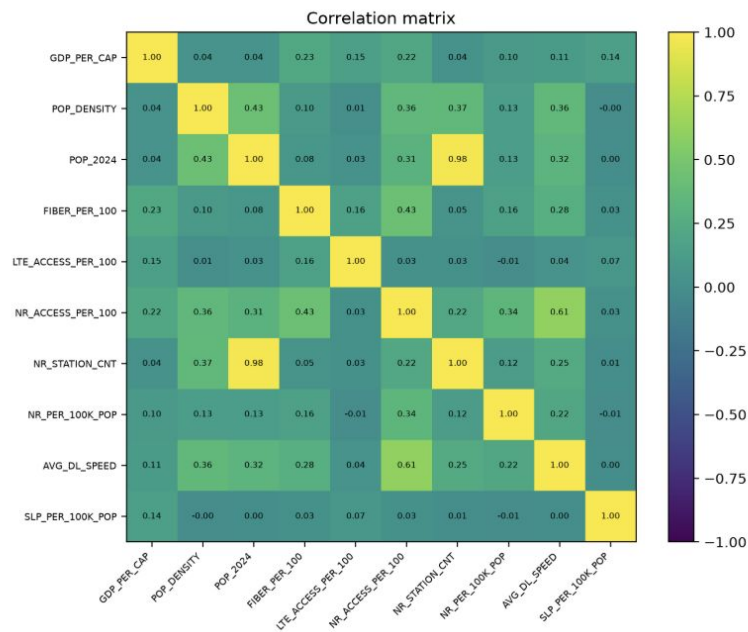
Skewed variables on raw versus log1p scales



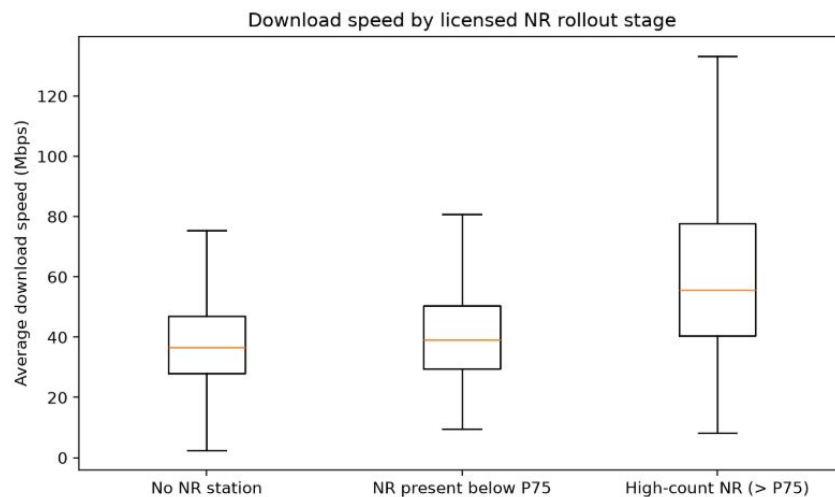
Note. Top row: raw scales. Bottom row: log1p scales. The zero spike in NR station counts reflects the 60.7% of municipalities with no NR station.

Data Description and EDA

Correlation Matrix for Modeled Variables



Download Speed by NR Rollout Stage

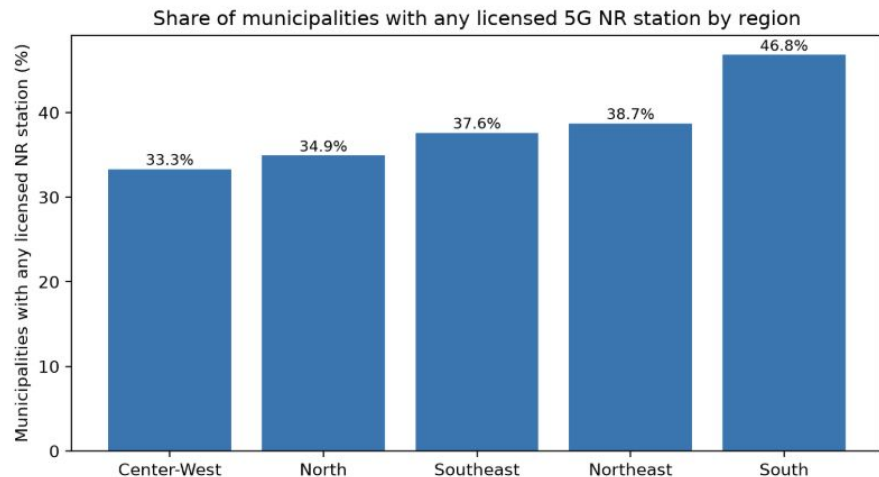


Note. Boxes show interquartile ranges without outliers.

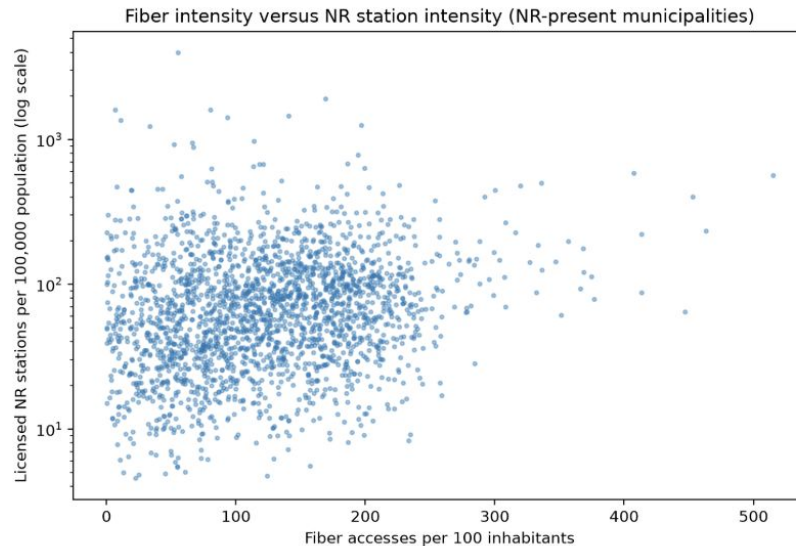
Data Description and EDA



Share of Municipalities With Any 5G NR Deployment by Region

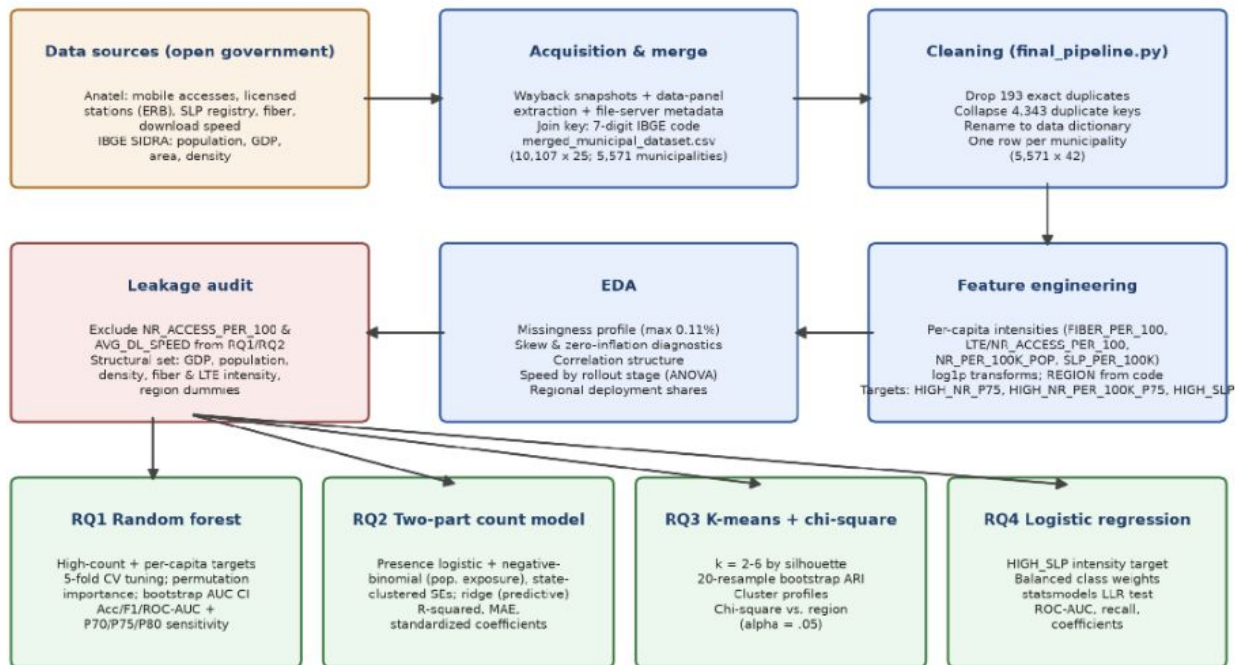


Fiber Intensity Versus NR Station Density (Log-Scaled)



Architecture diagram/Workflow

End-to-End Workflow of the Proposed System



Outputs: reports/figures, reports/tables, headline_results.json (fixed seed = 42; single command: `python final_pipeline.py`)

Model building



01

Leakage Remediation

A central methodological safeguard was the exclusion of "near-target" predictors—specifically, NR accesses per 100 and average download speeds. These variables were excluded because they are downstream consequences of 5G capacity and would artificially inflate performance.

02

Validation Protocol

Supervised models utilized a stratified 75/25 development split and repeated 5-fold cross-validation. To address spatial dependence, the study employed state-grouped cross-validation (GroupKFold).

03

Clustered Errors

Because municipalities within a single state share the same operators, regulations, and economic contexts, all inferential models reported state-clustered robust standard errors to provide accurate coefficient uncertainty.

Model building



RQ1: High-Count NR Rollout Classification (Random Forest)

Model Choice

A Random Forest classifier was used to predict whether a municipality qualifies as a high-count 5G NR rollout site.

Justification

Random forests were selected because they effectively handle non-linear classification and skewed predictors.

The model was deliberately restricted to a leakage-free set of purely structural predictors (population, density, GDP per capita, fiber intensity, and legacy LTE access) to measure true structural forecasting power rather than simply using downstream metrics that restate the target.

Model building



RQ2: Structural Correlates of NR Station Intensity (Hurdle Model)

Model Choice

A two-part hurdle model was implemented, consisting of a logistic regression (to predict the presence of any NR station) and a zero-truncated negative binomial model with a population offset (to predict station intensity among municipalities where NR is already present).

Justification

This two-part model was justified because the data was both highly overdispersed and heavily zero-inflated, with 60.7% of municipalities having exactly zero NR stations.

Theoretically, it correctly models the reality that a carrier's decision to enter a municipality (the extensive margin) is an entirely different investment process than deciding how many stations to deploy once entered (the intensive margin).

Model building



RQ3: Digital-Infrastructure Profile Segmentation (K-Means Clustering)

Model Choice

K-means clustering was used to group municipalities into digital-infrastructure profiles, with the optimal number of clusters selected via the silhouette coefficient.

Justification

K-means provided a transparent and reproducible method for partitioning the six-indicator feature space.

The choice of clusters was strictly descriptive to summarize infrastructure capacity, and was validated using bootstrap adjusted Rand indices for stability.

Model building



RQ4: Correlates of SLP Station Intensity (Balanced Logistic Regression)

Model Choice

A class-weighted, balanced logistic regression was built to predict the likelihood of high private-network-related SLP (Serviço Limitado Privado) station intensity.

Justification

Because the base SLP presence flag was saturated (present in nearly 98% of locations), the model targeted the 75th percentile of intensity instead.

A class-weighted approach was chosen specifically to favor recall (sensitivity) over specificity, ensuring the model acts as an effective exploratory screening signal for enterprise private-network feasibility.

Results



RQ1: High-Count NR Rollout

Result: A leakage-free random forest trained strictly on structural characteristics accurately discriminated high-count NR municipalities (ROC-AUC = .927, 95% CI [.906, .945]). [cite: 24]

Inference: Much of this predictive power is driven by municipal scale. Re-evaluated against a per-capita target, performance dropped to ~ROC-AUC .82, indicating population size is the dominant driver of the initial signal. [cite: 24]

Note: See Figure 11 (RQ1 Permutation Importance) for visualization details. [cite: 24]

RQ2: NR Station Intensity

Result: Predicting 5G presence (ROC-AUC = .849) is entirely different from predicting its intensity. The second stage analyzing continuous per-capita intensity yielded very weak results with no true predictive value. [cite: 24]

Inference: Within 5G-deployed municipalities, structural correlates have modest positive association. Station counts grow sub-proportionally with population ($\gamma = -.14$), meaning intensity declines as size increases. [cite: 24]

Results



RQ3: Digital-Infrastructure Profile

Result: K-means clustering ($k=2$) identified two stable but only softly separated infrastructure profiles (silhouette = .305). A Chi-square test confirmed a statistically significant association between these clusters and macro-regions ($p < .001$), though Cramér's V (.21) proved the association is relatively modest.

Inference: While regional inequality and inherited infrastructure clearly persist (with the higher-profile cluster exhibiting superior GDP, density, fiber, and speeds), these divides are not absolute. The modest regional association proves that geography alone does not fully dictate a digital readiness.

RQ4: SLP Station Intensity (Exploratory)

Result: A balanced logistic regression modeling high private-network-related SLP station intensity achieved a held-out ROC-AUC of .810. Odds ratios indicated significant positive associations with GDP per capita population LTE access, and fiber density.

Inference: Early indicators suggest enterprise private-network activity favors wealthier municipalities with legacy telecom infrastructure. However, performance dropped substantially when grouping cross-validation by state, highlighting high geographic heterogeneity.

Implementation and User Benefit



Deployment Accessibility

The framework is packaged as a reproducible Python pipeline (**final_pipeline.py**) that executes on standard CPU compute, allowing single-command regeneration.

System Integration

Outputs are standard CSV and JSON files keyed by the 7-digit IBGE code, enabling direct integration into GIS, BI dashboards, and planning systems.

Carrier Planning Efficiency

Ranked probability outputs reduce broad-scale prospecting by prioritizing structural shortlists rather than manual analysis of 5,571 municipalities.

Regulatory Monitoring

Anatel and MiniCom can predict and generate "watchlists" of structurally high-likelihood 5G areas reporting zero stations to guide inquiries.

Enterprise Benchmarking

Digital-infrastructure profiles provide comparative readiness benchmarks for private networks (SLP) before costly feasibility studies.

Conclusion



Infrastructure Asymmetry

The study successfully established a clear asymmetry in the deployment of 5G infrastructure in Brazil: while demographic, economic, and inherited telecommunication characteristics strongly predict *where* 5G appears and the raw count of stations, they explain significantly less of the per-capita *intensity* of those stations once deployed.

Predictive Limits

Predicting the presence of 5G infrastructure is possible (ROC-AUC = .849), but modeling continuous conditional intensity yields very low explanatory power, meaning structural municipal data cannot effectively forecast the engineering or strategic intensity of 5G investment.

Research Value

The primary value of this research is its creation of a disciplined, leakage-audited screening framework that separates scale from intensity, providing actionable prioritization inputs for regulators, carriers, and enterprises without claiming causal determinants or regulatory compliance.

Limitations



Construct Validity

The use of licensed NR and SLP stations serves as indirect footprints and does not actually confirm 5G Standalone operation, cloud-native architecture, or verified private 5G deployment.

Scale Conflation

The primary target variable (high count) inherently encodes municipal scale, which, despite mitigation efforts with per-capita metrics, means the two models ultimately answer slightly different questions.

Methodological Bounds

The predictors used in the study are structural correlates rather than exogenous causes, and spatial dependence remains a factor even after applying state-grouped validation methods.

Data Provenance

Upstream provenance is incomplete; while source families are identified, the historical raw snapshots and exact historical grains for the SLP and measured-speed data remain unresolved.

Future Work



Data Archiving

To improve accuracy, future studies should reconstruct the merge from newly extracted Anatel data and strictly archive all raw files, checksums, and timestamps locally.

Methodological Expansion

Transitioning from a cross-sectional approach to a longitudinal panel tracking data around auction-obligation dates would allow researchers to observe change over time and incorporate spatial models.

Refined Variables

Construct validity could be greatly enhanced by utilizing variables that directly measure Standalone (SA) and Non-Standalone (NSA) status, virtualized RAN metrics, spectrum band details, enterprise density, and actual deployment costs.

Bibliography



- Agência Nacional de Telecomunicações. (2021). *Leilão de espectro 5G* [5G spectrum auction]. <https://www.gov.br/anatel/pt-br/assuntos/5G/leilao-de-espectro-5g>
- Agência Nacional de Telecomunicações. (2026a). *Painéis de dados da Anatel* [Anatel data panels]. <https://informacoes.anatel.gov.br/paineis/>
- Agência Nacional de Telecomunicações. (2026b). *Dados abertos da Anatel* [Anatel open data]. <https://www.anatel.gov.br/dadosabertos/>
- Agência Nacional de Telecomunicações. (2026c). *Inventário de bases de dados* [Database inventory] [Data set]. https://www.anatel.gov.br/dadosabertos/PDA/Bases_Publicadas/Inventario_de_Bases_de_Dados.csv
- Abrardi, L., & Cambini, C. (2019). Ultra-fast broadband investment and adoption: A survey. *Telecommunications Policy*, 43(3), 183–198. <https://doi.org/10.1016/j.telpol.2019.02.005>
- Barrachina-Muñoz, S., Payaró, M., & Mangles-Bafalluy, J. (2022). Cloud-native 5G experimental platform with over-the-air transmissions and end-to-end monitoring. In *2022 13th International Symposium on Communication Systems, Networks and Digital Signal Processing (CSNDSP)* (pp. 22–26). IEEE. <https://doi.org/10.1109/CSNDSP54353.2022.9908028>
- Bauer, J. M., & Bohlin, E. (2022). Regulation and innovation in 5G markets. *Telecommunications Policy*, 46(4), Article 102260. <https://doi.org/10.1016/j.telpol.2021.102260>
- Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32. <https://doi.org/10.1023/A:1010933404324>

Bibliography



Cameron, A. C., & Trivedi, P. K. (2013). *Regression analysis of count data* (2nd ed.). Cambridge University Press.

Cohen, J. (1988). *Statistical power analysis for the behavioral sciences* (2nd ed.). Lawrence Erlbaum Associates.

Cramér, H. (1946). *Mathematical methods of statistics*. Princeton University Press.

Ericsson. (2025). *Ericsson mobility report: June 2025*. <https://www.ericsson.com/en/reports-and-papers/mobility-report/reports/june-2025>

Gozzi, N., Comini, N., & Perra, N. (2024). Bridging the digital divide: Mapping Internet connectivity evolution, inequalities, and resilience in six Brazilian cities. *EPJ Data Science*, 13, Article 69. <https://doi.org/10.1140/epjds/s13688-024-00508-8>

Green, S. B. (1991). How many subjects does it take to do a regression analysis? *Multivariate Behavioral Research*, 26(3), 499–510. https://doi.org/10.1207/s15327906mbr2603_7

GSMA. (2024). *The mobile economy Latin America 2024*. GSMA Intelligence. <https://www.gsma.com/solutions-and-impact/connectivity-for-good/mobile-economy/latam-2024/>

Hubert, L., & Arabie, P. (1985). Comparing partitions. *Journal of Classification*, 2(1), 193–218. <https://doi.org/10.1007/BF01908075>

Instituto Brasileiro de Geografia e Estatística. (2024). *Síntese de indicadores sociais: Uma análise das condições de vida da população brasileira: 2024* [Synthesis of social indicators].

<https://www.ibge.gov.br/estatisticas/sociais/populacao/9221-sintese-de-indicadores-sociais.html>

Bibliography



- Instituto Brasileiro de Geografia e Estatística. (2025). *Produto interno bruto dos municípios: Brasil 2022–2023: Notas técnicas* [Gross domestic product of municipalities: Brazil 2022–2023: Technical notes]. IBGE. <https://www.ibge.gov.br/estatisticas/economicas/contas-nacionais/9088-produto-interno-bruto-dos-municipios.html>
- Instituto Brasileiro de Geografia e Estatística. (2026). *API de agregados – SIDRA* [Aggregates API – SIDRA]. <https://servicodados.ibge.gov.br/api/docs/agregados>
- International Telecommunication Union. (2017). *Recommendation ITU-T Y.3100: Terms and definitions for IMT-2020 network*. <https://www.itu.int/rec/T-REC-Y.3100/en>
- Kaufman, S., Rosset, S., Perlich, C., & Stitelman, O. (2012). Leakage in data mining: Formulation, detection, and avoidance. *ACM Transactions on Knowledge Discovery from Data*, 6(4), Article 15. <https://doi.org/10.1145/2382577.2382579>
- Knieps, G., & Bauer, J. M. (2022). Internet of things and the economics of 5G-based local industrial networks. *Telecommunications Policy*, 46(4), Article 102261. <https://doi.org/10.1016/j.telpol.2021.102261>
- Lambert, D. (1992). Zero-inflated Poisson regression, with an application to defects in manufacturing. *Technometrics*, 34(1), 1–14. <https://doi.org/10.2307/1269547>
- Macedo, H. R., & Carvalho, A. X. Y. (2013). Análise de possíveis determinantes da penetração do serviço de acesso à internet em banda larga nos municípios brasileiros [An analysis of the possible determinants of broadband internet access penetration in Brazilian municipalities]. *Economia e Sociedade*, 22(2), 483–520. <https://www.scielo.br/j/ecos/a/KS3SKRGmpT6j4tfbqhggjGGy/>

Bibliography



- Mullahy, J. (1986). Specification and testing of some modified count data models. *Journal of Econometrics*, 33(3), 341–365. [https://doi.org/10.1016/0304-4076\(86\)90002-3](https://doi.org/10.1016/0304-4076(86)90002-3)
- Nishijima, M., Ivanauskas, T. M., & Sarti, F. M. (2017). Evolution and determinants of digital divide in Brazil (2005–2013). *Telecommunications Policy*, 41(1), 12–24. <https://doi.org/10.1016/j.telpol.2016.10.004>
- Nokia. (2025). *Nokia partners with Nvidia*. <https://www.nokia.com/newsroom/nokia-partners-with-nvidia/>
- O-RAN ALLIANCE. (2026). *O-RAN specifications*. <https://www.o-ran.org/specifications>
- Organisation for Economic Co-operation and Development. (2020). *OECD telecommunication and broadcasting review of Brazil 2020*. OECD Publishing.
https://www.oecd.org/en/publications/oecd-telecommunication-and-broadcasting-review-of-brazil-2020_30ab8568-en.html
- Organisation for Economic Co-operation and Development. (2022). *Developments in spectrum management for communication services* (OECD Digital Economy Papers No. 332). OECD Publishing.
https://www.oecd.org/content/dam/oecd/en/publications/reports/2022/10/developments-in-spectrum-management-for-communication-services_ae18e03f/175e7ce5-en.pdf
- Organisation for Economic Co-operation and Development. (2025). *Closing broadband connectivity divides for all*. OECD Publishing.
https://www.oecd.org/en/publications/closing-broadband-connectivity-divides-for-all_d5ea99b2-en.html

Bibliography



Oughton, E. J., & Frias, Z. (2018). The cost, coverage and rollout implications of 5G infrastructure in Britain. *Telecommunications Policy*, 42(8), 636–652.

<https://doi.org/10.1016/j.telpol.2017.07.009>

Polese, M., Bonati, L., D'Oro, S., Basagni, S., & Melodia, T. (2023). Understanding O-RAN: Architecture, interfaces, algorithms, security, and research challenges. *IEEE Communications Surveys & Tutorials*, 25(2), 1376–1411. <https://doi.org/10.1109/COMST.2023.3239220>

Roberts, D. R., Bahn, V., Ciuti, S., Boyce, M. S., Elith, J., Guillerá-Arroita, G., Hauenstein, S., Lahoz-Monfort, J. J., Schröder, B., Thuiller, W., Warton, D. I., Wintle, B. A., Hartig, F., & Dormann, C. F. (2017). Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography*, 40(8), 913–929. <https://doi.org/10.1111/ecog.02881>

Rousseeuw, P. J. (1987). Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. *Journal of Computational and Applied Mathematics*, 20, 53–65.

[https://doi.org/10.1016/0377-0427\(87\)90125-7](https://doi.org/10.1016/0377-0427(87)90125-7)

The Internet Archive. (2026). *Wayback Machine*. <https://web.archive.org/>

Bibliography



Instituto Brasileiro de Geografia e Estatística. (2026a). Produto interno bruto dos municípios: Tabela 5938 [Data set]. SIDRA. <https://sidra.ibge.gov.br/tabela/5938>

Instituto Brasileiro de Geografia e Estatística. (2024). População residente estimada: Tabela 6579 [Data set]. SIDRA. <https://sidra.ibge.gov.br/tabela/6579>

Instituto Brasileiro de Geografia e Estatística. (2022). População residente, área territorial e densidade demográfica: Tabela 4714 [Data set]. SIDRA. <https://sidra.ibge.gov.br/tabela/4714>

Agência Nacional de Telecomunicações. (2026h). Medidas de qualidade da telefonia móvel [Data set]. <https://informacoes.anatel.gov.br/paineis/>

Agência Nacional de Telecomunicações. (2026g). Redes privadas [Data set]. <https://www.gov.br/anatel/pt-br/regulado/radiofrequencia/redes-privativas>

Agência Nacional de Telecomunicações. (2026f). Meu Município: Acessos e cobertura [Data set]. <https://informacoes.anatel.gov.br/paineis/>

Agência Nacional de Telecomunicações. (2026e). Estações licenciadas [Data set]. https://www.anatel.gov.br/dadosabertos/paineis_de_dados/outorga_e_licenciamento/estacoes_licenciadas.zip

Agência Nacional de Telecomunicações. (2026d). Acessos autorizadas SMP [Data set]. https://www.anatel.gov.br/dadosabertos/paineis_de_dados/acessos/acessos_telefonia_movel.zip